BMS COLLEGE OF ENGINEERING

(Autonomous college, affiliated to Visveswaraya Technological University)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION

V Semester

ADVANCED DIGITAL LOGIC DESIGN

(23EC5PE1AD)

AAT Report-2

Topic: Digital Logic Modules Simulations

Sl No	Student Name	USN	Student signature	Signature of the Staff In charge	Marks Obtained
1.	NEHAL G D	1BF24EC090			

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Github link-https://github.com/nehal-gd/1BF24EC090-ADLD/upload/main

Tools Used

The design, implementation, and verification of the twelve digital logic modules described in this report were conducted using an industry-standard Electronic Design Automation (EDA) software suite running within an enterprise Linux environment. The complete simulation and debugging flow relies on Cadence Design Systems software tools, providing high-performance compilation, simulation, and waveform analysis capabilities.

Linux Operating System Environment

All EDA tools were deployed on a 64-bit Linux distribution, which serves as the standard operating system environment for semiconductor design and verification in enterprise and academic settings. The Linux environment provides stable process management, efficient file system handling, and robust multi-user resource allocation required for high-throughput digital logic simulations and command-line automation via shell scripting.

Cadence Xcelium / NC-Verilog Simulator

Logic compilation and dynamic functional simulation were performed using the Cadence logic simulation engine (NC-Verilog / Xcelium Enterprise Simulator). The compiler processes IEEE 1364-compliant Verilog HDL module descriptions alongside their corresponding testbenches. The simulator executes event-driven functional verification, enforcing precise timing control (`timescale 1ns/1ps), evaluation of logic gate delays, and signal transitions across all internal design nodes.

Cadence SimVision Waveform Viewer

Waveform analysis and visual signal verification were conducted using Cadence SimVision. During simulation execution, value change dump files (.vcd) or database files were generated and imported into SimVision to inspect time-domain logic behavior. The graphical interface enables precise timing measurements, bus signal expansion, state machine transition tracking, and glitch analysis across test vectors to validate full design correctness prior to hardware synthesis.

1.Divide by 'n' counter

A Divide-by-N counter is a sequential logic circuit that divides the input clock frequency by a fixed integer value N

It counts a specific number of clock pulses and then resets or repeats its counting sequence. The output frequency is equal to the input frequency divided by N

Verilog-RTL and TESTBENCH CODE

*div_n_counter.v (~/cadWorkDir/Digital/1BF24EC090/SIM/RTL)

File Edit View Search Tools Documents Help

```
module div_n_counter(input clk,input reset,output reg [1:0]q);

always@(posedge clk or posedge reset)
begin
    if(reset)
    q<= 2'b00;
    else
    q<=q+1;
end

endmodule
```

tb_counter.v (~/cadWorkDir/Digital/1BF24EC090/SIM/TB)

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tb_comp_2.v × tb_counter.v ×

```
module tb_counter();

reg clk,reset;
wire [1:0]q;

div_n_counter  DUT(.clk(clk),.reset(reset),.q(q));

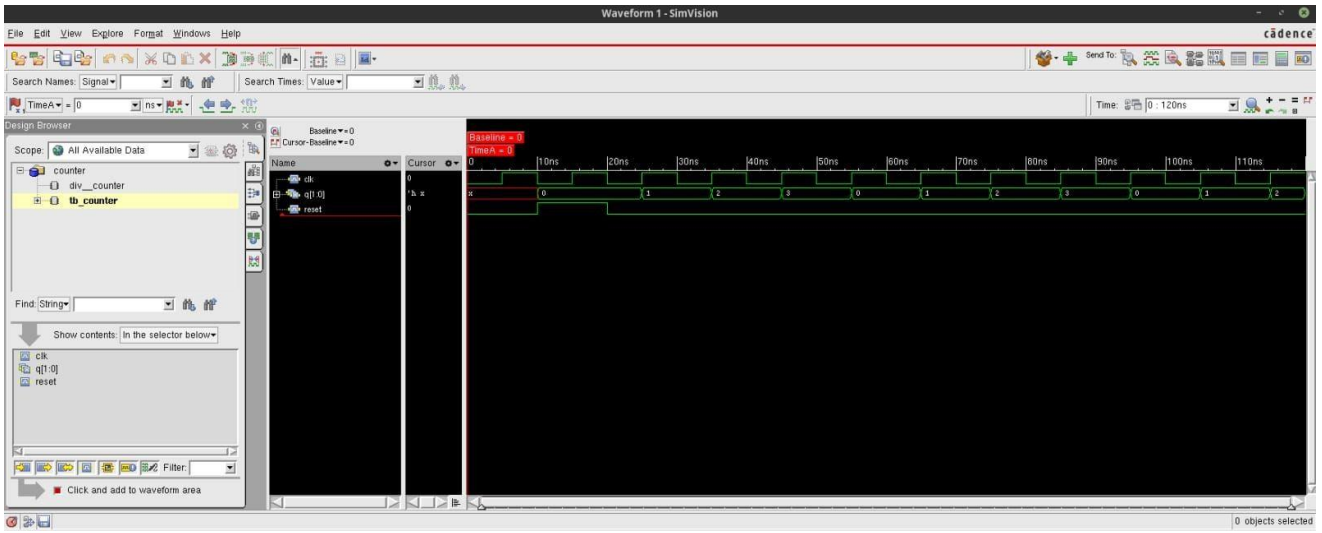
initial
begin
clk = 0;
reset=0;
#10
reset=1;
#10
reset=0;
#100
$finish;
end

initial begin
$dumpfile("counter.vcd");
$dumpvars;
end

always #5 clk = ~clk;

endmodule
```

Simulation-



2. 4-bit Comparator

A 4-bit comparator is a combinational logic circuit used to compare two 4-bit binary numbers and determine their relative magnitude

It compares two binary inputs and produces three outputs:
1. A > B – Indicates that A is greater than B
2. A = B – Indicates that A is equal to B
3. A < B – Indicates that A is less than B

Verilog-RTL and TESTBENCH CODE

comp_2.v (~/cadWorkDir/Digital/1BF24EC090/SIM/RTL)

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*div_n_counter.v × comp_2.v ×

```
`timescale 1ns/1ps

// 4-bit Cascadable Magnitude Comparator

module comp_2(
    input  [3:0] a, b,
    input        gt_in, eq_in, lt_in,
    output       gt, eq, lt
);

    wire [3:0] x;
    wire       eq_w;
    wire       gt_w;
    wire       lt_w;

    // XNOR operation
    assign x = ~(a ^ b);

    // Equality logic
    assign eq_w = &x;

    // Greater than logic
    assign gt_w = (a[3] & ~b[3])
                | (x[3] & a[2] & ~b[2])
                | (x[3] & x[2] & a[1] & ~b[1])
                | (x[3] & x[2] & x[1] & a[0] & ~b[0]);

    // Less than logic
    assign lt_w = (~a[3] & b[3])
                | (x[3] & ~a[2] & b[2])
                | (x[3] & x[2] & ~a[1] & b[1])
                | (x[3] & x[2] & x[1] & ~a[0] & b[0]);

    // Cascading output logic
    assign gt = gt_w | (eq_w & gt_in);
    assign eq = eq_w & eq_in;
    assign lt = lt_w | (eq_w & lt_in);

endmodule
```

tb_comp_2.v (~/cadWorkDir/Digital/1BF24EC090/SIM/TB)

File Edit View Search Tools Documents Help

```
timescale 1ns/1ps

module tb_comp_2;

    reg  [3:0] a, b;
    reg        gt_in, eq_in, lt_in;
    wire       gt, eq, lt;

    // Design Under Test
    comp_2 DUT(
        .a(a),
        .b(b),
        .gt_in(gt_in),
        .eq_in(eq_in),
        .lt_in(lt_in),
        .gt(gt),
        .eq(eq),
        .lt(lt)
    );

    initial begin

        $dumpfile("dump.vcd");
        $dumpvars(0, tb_comp_2);

        $monitor("Time=%0t ns | A=%d B=%d | GT=%b EQ=%b LT=%b",
                 $time, a, b, gt, eq, lt);

        // Stand-alone comparator inputs
        gt_in = 1'b0;
        eq_in = 1'b1;
        lt_in = 1'b0;

        // Test 1: A > B
        a = 4'd5;  b = 4'd3;   #10;

        // Test 2: A < B
        a = 4'd3;  b = 4'd5;   #10;

        // Test 3: A = B
        a = 4'd7;  b = 4'd7;   #10;

        // Test 4: A > B
        a = 4'd8;  b = 4'd7;   #10;

        // Test 5: A < B
        a = 4'd0;  b = 4'd15;  #10;

        // Test 6: A = B
        a = 4'd15; b = 4'd15;  #10;

        // Test 7: A = B
        a = 4'd0;  b = 4'd0;   #10;

        // Test 8: A > B
        a = 4'd12; b = 4'd4;   #10;

        // Test 9: A < B
        a = 4'd2;  b = 4'd10;  #10;

        $finish;

    end
```

Simulation-

